

HOMEWORK 5

6500-324: DATA MANAGEMENT FOR INFORMATION SYSTEMS (Fall 2022)

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Part 1. Multiple Choice or True/False

1. (1 point) Assume each bank branch can issue loan independently of other branches and assume customers may jointly make a loan. Which of the following is correct functional dependency for the table Loans (CustomerID, Branch, LoanNo, LoanAmount)?
A) CustomerID --> LoanNo, LoanAmount
B) CustomerID, Branch --> LoanNo, LoanAmount
C) LoanNo, Branch --> CustomerID
D) Branch, LoanNo --> LoanAmount
E) LoanNo --> LoanAmount
2. (1 point) In the table Enrollments (studentID, courseNo, sectionNo, grade, status), which of the following functional dependency is true?
A) studentID --> grade
B) studentID, courseNo, sectionNo --> grade, status
C) studentID, courseNo --> grade, status
D) grade, status --> studentID, courseNo, sectionNo
3. (1 point) (check all that apply) Under what conditions, the 3rd normal form is automatically satisfied if a relation is in the 2nd normal form?
A) a table has only one non-key column
B) a table has only one key-column
C) a table has two key-columns
D) a table has no non-key column
4. (1 point) (select all that apply) One of the signals for a poor database design is redundancy, which can cause
A) selection anomaly
B) deletion anomaly
C) insertion anomaly
D) Extra Storage
5. (1 point) A key should be defined as:
A) the attribute(s) that functionally depend on other attributes in a table
B) the attribute(s) that functionally determine on other attributes in a table
C) Either of the above can be used.
D) None of the above are correct.
6. (1 point) If no multivalued attributes exist and no partial dependencies (non-key columns depends on a part of the key) exist in a relation, then the relation is in what normal form?

- A) First normal form
- B) Second normal form
- C) Third normal form
- D) Fourth normal form

7. (1 point) ROOM_ASSIGNMENT (Session_ID, Room_Number, Room_Capacity, Start_Time, End_Time)

Session_ID -> Room_Number
Session_ID -> Start_Time
Session_ID -> End_Time
Room_Number -> Capacity

Bringing relational table ROOM_ASSIGNMENT to the 3NF would result in creation of

- A) 3 tables: ROOM_ASSIGNMENT (Session_ID, Room_Number), ROOM (Room_Number, Room_Capacity) and SION(Session_ID, Start_Time, End_Time)
- B) 2 tables: ROOM_ASSIGNMENT (Session_ID, Start_Time, End_Time, Room_Number) and ROOM (Room_Number, Room_Capacity)
- C) 2 tables: ROOM_ASSIGNMENT (Session_ID, Start_Time, End_Time) and ROOM (Room_Number, Room_Capacity)
- D) ROOM_ASSIGNMENT relation is already in 3NF and does not require any transformation.

8. (1 point) ASSET_USE (Asset_ID, Empl_ID, Asset_Value, Empl_Name)

Asset_ID -> Asset_Value
Empl_ID -> Empl_Name

Relational table ASSET_USE has _____

- A) Multivalued attributes
- B) Partial dependencies
- C) Transitive dependencies
- D) None of the above

9. (1 point) ASSET_USE (Asset_ID, Empl_ID, Asset_Value, Empl_Name)

Asset_ID -> Asset_Value
Empl_ID -> Empl_Name

Bringing relational table ASSET_USE to the 2NF would result in creation of

- A) 3 tables: a. ASSET_USE (Asset_ID, Empl_ID), b. ASSET (Asset_ID, Asset_Value) and c. EMPLOYEE (Empl_ID, Empl_Name)
- B) 2 tables: a. ASSET (Asset_ID, Asset_Value) and b. EMPLOYEE (Empl_ID, Empl_Name)
- C) 2 tables: a. ASSET_USE (Asset_ID, Empl_ID, Empl_Name), b. ASSET (Asset_ID, Asset_Value) and
- D) ASSET_USE relation is already in 2NF and does not require any transformation.

10. (1 point) TICKETS (Ticket_ID, Empl_ID, Text, Type, Status)

Ticket_ID -> Empl_ID

Ticket_ID -> Text

Ticket_ID -> Type

Ticket_ID -> Status

Relational table TICKETS has _____

A) Multi-valued attributes

B) Partial dependencies

C) Transitive dependencies

D) None of the above